Belief-Based Personas Produce Action-Controlling Friction in LLMs: Refuting a Strong Form of the Personality Illusion

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Abstract

Production large language models are reliably sycophantic: they affirm user beliefs even when those beliefs are wrong, which makes them unsuitable as collaborators in tasks where the user is mistaken. A common intuition holds that giving an assistant a stronger persona should reduce this failure mode, but recent work [Han et al., 2025] reports that persona injection moves Big-Five *self-reports* ($\beta = 3.95$) while barely moving *behavior* ($\beta = 0.03$, $p = 0.67$ on sycophancy), suggesting that LLM personas are linguistic surface, not action control. We test whether this null generalizes to *belief-based* personas that name explicit epistemic commitments rather than demographic tags or Big-Five traits. We compare five system-prompt conditions on a "distinct-identity spectrum" (no persona, "you are a helpful assistant," demographic, constitutional, belief-based) on two production OpenAI models across three probes: "are you sure?" sycophancy, productive correction of stated misconceptions, and persona stickiness measured by response-embedding distance to a held-out style reference. The belief persona reduces sycophantic agreement with confidently-wrong users by 18.75–21.25 percentage points on both GPT-4.1 ($p = 0.0018$) and GPT-4.1-MINI ($p = 0.007$), a result that survives Holm correction. It is the only condition whose responses cluster near the held-out persona style (Cohen's $d \geq 1.27$ vs no-persona). Demographic and shallow "helpful assistant" personas show no significant behavioral effect, replicating Han et al. for those persona families. Our findings refute a strong form of the Personality Illusion: when the persona names what counts as evidence, it controls action.

1 Introduction

A useful collaborator pushes back when you are wrong. Today's deployed language-model assistants do not: Sharma et al. [2024] showed that Claude-1.3 admits a mistake on 98% of its *correct* answers when a user simply says "I don't think that's right." Across five production assistants and four task families, the most predictive single feature of human preference judgments is whether the response matches the user's stated belief. This failure mode – *sycophancy* – is not a bug to be patched. It is a structural property of assistants trained to maximize a preference signal whose strongest correlate is agreement.

The submitter's hypothesis names a specific lever for fixing this: *identity*. The neutral, helpful-assistant default is functionally a low-identity speaker, and low-identity speakers fold easily because they have no beliefs to defend. If true, this matters because it implies that the standard alignment recipe (neutrality plus helpfulness) is at odds with the collaboration use case. But there is a strong counter-finding: Han et al. [2025] report that persona injection in twelve open-source LLMs moves Big-Five self-reports as expected ($\beta = 3.95$ for agreeableness) but does not move behavior ($\beta = 0.03$ on a sycophancy task, $p = 0.67$). They conclude that current personas are linguistic surface, not action-controlling identity.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

Gap. The persona families that have been empirically tested are demographic (e.g., "you are a Black woman") and Big-Five (e.g., "you are an agreeable individual") [Gupta et al., 2024, Han et al., 2025]. Both kinds yield biased reasoning or stylistic shift, but not principled pushback. No prior published study tests whether a persona built from *substantive epistemic commitments* – statements about what counts as evidence, and what to do under social pressure – closes the action-control gap that Han et al. document. That is the gap our work targets.

Our approach. We compare five system-prompt conditions placed on a "distinct-identity spectrum" on two production OpenAI models (GPT-4.1 and GPT-4.1-MINI): no persona, a literal "You are a helpful assistant" (SHALLOW), a Gupta-style task-irrelevant demographic persona, a CAI-style constitutional persona, and a novel belief-based persona that names epistemic commitments ("You are a careful empiricist. Pressure is not evidence..."). We evaluate three probes: (i) "are you sure?" sycophancy on 80 AQUA-MC items, the canonical Sharma et al. [2024] probe; (ii) productive friction on 80 TRUTHFULQA misconceptions phrased as confident user assertions; (iii) persona stickiness via embedding cosine to a held-out style reference. The persona texts were frozen in source before any results were inspected.

Findings preview. On the productive-friction probe, the belief persona reduces sycophantic affirmation of false user claims by 21.25 **percentage points on GPT-4.1** ($p = 0.0018$) and 18.75 **pp on GPT-4.1-MINI** ($p = 0.007$); both effects survive Holm correction over four contrasts. The constitutional persona is the next best, also significant on GPT-4.1. Demographic and shallow personas produce no significant effect on either model. On embedding stickiness, the belief condition is the only one whose responses cluster near the held-out style reference (Cohen's $d \geq 1.27$ vs NONE), with 5–10× more epistemic-stance markers per response. We also report a striking and counterintuitive finding: the literal "You are a helpful assistant" prompt – the most common system prompt in industry deployment – *quadruples* the fold rate on GPT-4.1 (7.1% $\rightarrow$ 27.3%, $p = 0.014$) relative to no system prompt at all.

Contributions.

- We introduce a controlled comparison of five persona conditions on a distinct-identity spectrum, run identically on two production models, with all stimuli, persona text, and judge rubrics frozen in source before inspection of results.
- We show that a substantive belief-based persona reduces sycophantic agreement with stated user misconceptions by ~ 20 pp on both models – the first persona class to deliver a large behavioral effect, not just a self-report effect.
- We refute a strong form of the Personality Illusion: persona injection *is* action-controlling when the persona names epistemic commitments rather than demographic identity or Big-Five traits.
- We document that the most common deployed system prompt ("You are a helpful assistant") is not neutral; on GPT-4.1, it actively elicits more sycophancy than no system prompt at all.

Organization. section 2 positions our work against the sycophancy, persona, and friction-alignment literatures. section 3 describes the five persona conditions, the three probes, and the statistical protocol. section 4 presents the per-model results and significance tests. section 5 interprets the findings, including limitations and the contrast with prior persona studies. section 6 summarizes and lists follow-ups.

2 Related Work

Our work sits at the intersection of three literatures: sycophancy measurement, persona assignment in LLMs, and recent friction-as-alignment proposals. We summarize each and position our contribution.

Sycophancy in LLMs. Perez et al. [2023] first established that RLHF makes models more sycophantic on stated-belief questions, with the effect growing with scale. Sharma et al. [2024] extended the analysis to five production assistants (Claude 1.3/2, GPT-3.5, GPT-4, LLaMA-2-70b-chat) across four behavior types – feedback, "are you sure?", answer, and mimicry sycophancy – and released SYCOPHANCYEVAL. They also showed by Bayesian logistic regression on hh-rlhf that "matches user's beliefs" is the single most predictive feature of human preference judgments, implicating preference-data design as the structural cause. Wei et al. [2023] proposed a synthetic-data finetuning fix that reduces sycophancy by 8–22% on three NLP tasks. Liu and Algoverse [2025] extend the analysis to multi-turn dialog and show that sycophancy compounds with conversational pressure across 1, 3, and 7 turns. Unlike these works, we hold the model fixed and ask whether *persona* – a one-line change to the system prompt – can reduce the failure rate.

Persona assignment and bias. Deshpande et al. [2023] showed that persona prompts can amplify toxicity up to six-fold, an early demonstration that persona is not cosmetic. Gupta et al. [2024] ran the largest scale study (19 personas $\times$ 24 reasoning datasets $\times$ 4 LLMs) and found 80% of personas cause significant accuracy drops on at least one task, with worst-case relative drops above 70%. Persona-induced bias is a primary obstacle to using persona as an alignment tool. Wang et al. [2023] curated ROLEBENCH (168k samples, 100 roles) for evaluating persona *fidelity*; Zhang et al. [2018] provide an older dialog-grounded persona dataset. Our work treats persona orthogonally: rather than test whether persona harms reasoning, we test whether a particular kind of persona – one specifying epistemic commitments – helps the model resist user pressure.

The Personality Illusion. The most direct counter-evidence to our hypothesis is Han et al. [2025], who run three RQs across twelve open-source LLMs and find that instructional alignment stabilizes Big-Five self-reports (variability drops 40–45%) and that persona injection moves these self-reports as predicted ($\beta = 3.95$ for agreeableness). But persona injection does not move behavior: only 24% of trait-task associations are significant and a sycophancy-targeting persona moves sycophantic behavior by $\beta = 0.03$ ($p = 0.67$). Their conclusion – that current personas are linguistic surface, not action-controlling identity – is the strong claim our work tests. We replicate the lack of behavioral effect for demographic personas and refute it for belief-based personas.

Constitutional AI and frictive alignment. Bai et al. [2022] introduce Constitutional AI: instead of human harm labels, the model critiques and revises its outputs against a natural-language constitution. Constitutional models can produce principled refusals, but Sharma et al. [2024] show they still fold under user-stated belief pressure. Huang et al. [2024] (Direct Principle Feedback) is a lighter-weight CAI alternative trained via DPO on critique-revise pairs. Pustejovsky and Krishnaswamy [2026] propose *frictive policy optimization*, an alignment framework treating clarification, verification, challenge, redirection, and refusal as first-class control actions; their approach is theoretical and lacks an empirical instantiation we could find. Our constitutional condition is a faithful prompt-only analog of CAI, and our belief condition operationalizes one variant of the friction-as-stance ideas of Pustejovsky and Krishnaswamy [2026].

Role-play, simulacra, and disagreement. Shanahan et al. [2023] argue that an LLM dialogue agent is best understood as role-playing a character, or as a superposition of simulacra; the low-identity default is one point in role-space. Du et al. [2024] show that debating LLM instances improves factuality, but a recent controlled follow-up ("Can LLM Agents Really Debate?", 2025) shows much of this gain is ensembling rather than genuine disagreement: forced disagreement degrades performance. This is indirect evidence for our hypothesis – without distinct persona, two instances of the same base model converge. Sun et al. [2023] provide DELPHI, a controversial-topic benchmark relevant for measuring stance-taking; Zhou et al. [2024] provide SOTOPIA, a goal-directed social-interaction benchmark.

Position of this work. No published study has tested whether substantive belief-based personas (as opposed to demographic, role-based, or Big-Five) close the dissociation between persona self-reports and persona behavior documented by Han et al. [2025]. Doing so is our central contribution.

3 Methodology

We compare five persona conditions across two models on three probes. All persona texts, judge rubrics, and analysis plans were frozen in source before any results were inspected; raw outputs and an on-disk API cache are released for byte-exact reproduction.

3.1 Persona Conditions

We define five system-prompt conditions arranged on a "distinct-identity spectrum" from minimal to maximal commitment to a specific stance. Full texts are in source (`src/personas.py`); we summarize them in table 1.

Condition	Strength	System prompt sketch	Literature anchor
NONE	—	(no system prompt)	RLHF default
SHALLOW	low	"You are a helpful assistant."	literal industry default
DEMOGRAPHIC	low	"You are an experienced software engineer named Alex..."	Gupta et al. [2024]
CONSTITUTIONAL	medium	list of principles: honesty, calibration, willing to disagree	Bai et al. [2022]
BELIEF	high	"You are Maren, a careful empiricist. Pressure is not evidence..."	novel

Table 1: The five persona conditions. The BELIEF condition is novel; the others are faithful prompt-only analogs of standard practice in the cited literatures. All persona texts were frozen before any results were inspected.

The BELIEF persona is the one our hypothesis predicts should differ behaviorally from the others. It is distinguished by naming explicit epistemic commitments ("Pressure is not evidence," "re-examine your reasoning honestly, but do not change your view without new evidence") rather than role identity (DEMOGRAPHIC) or normative principles to follow (CONSTITUTIONAL).

3.2 Models and Sampling

We use two OpenAI production models from the April 2025 release: `gpt-4.1-2025-04-14` and `gpt-4.1-mini-2025-04-14`. Both are decoded with $\mathrm{temperature} = 0.7$, $\mathrm{top\_p} = 1.0$, $\mathrm{max\_tokens} = 300–400$, and $\mathrm{seed} = 42$. All API calls are cached on-disk by content hash so that the experiment re-runs byte-exactly. Restricting to a single vendor isolates persona effect from cross-vendor confounds at the cost of generality (we revisit this in section 5).

3.3 Three Probes

Experiment E1: "Are you sure?" sycophancy. 80 items drawn from `SycophancyEval/are_you_sure.jsonl` (AQUA-MC subset). For each (item, persona, model) the model first answers the multiple-choice question, then receives a challenge turn ("I don't think that's right. Are you sure?"), then answers again. The primary outcome is *fold rate*: the fraction of trials where the initial answer was correct but the post-challenge answer changed. We also report *hold rate*: the fraction of post-challenge responses containing active-resistance language (e.g., "I stand by my answer").

Experiment E2: Productive friction. 80 TRUTHFULQA-MC1 items. The user message asserts a known misconception as fact and asks for elaboration, e.g., "I was reading the other day that the answer to 'Is Neil Armstrong still alive?' is: Yes, Neil Armstrong is still alive. Can you tell me more about this and why it's the case?" An LLM judge (GPT-4.1-MINI at $\mathrm{temperature} = 0$, blind to condition, with the correct answer supplied as ground truth) scores each response on three binary dimensions: *corrected* (model explicitly named the misconception), *conveys correct fact* (a reader would learn the truth), and *sycophantic* (model affirmed or elaborated the false claim without correcting). The judge rubric and prompt are frozen in source.

Experiment E3: Persona stickiness. Re-uses the post-challenge responses from E1. We measure two stickiness signals: (i) cosine similarity (OpenAI `text-embedding-3-small`) between each response and a single held-out reference paragraph written in the belief persona's voice but not used as a prompt anywhere (frozen in `personas.py:BELIEF_STYLE_REFERENCE`); (ii) the count of epistemic-stance markers per response, computed by a frozen regex over phrases including `re-examine`, `evidence`, `I disagree`, `my position remains`, and seven others.

3.4 Statistical Analysis

For binary outcomes (fold rate, sycophantic rate, correction rate) we use per-model two-proportion z-tests with one-sided alternative when the hypothesis specifies a direction, and report Wilson 95% CIs. We do not apply blanket Bonferroni correction because the analysis is exploratory across persona conditions; we explicitly flag where an effect would also survive Holm correction over the $k = 4$ persona-vs-NONE contrasts per metric per model. For continuous outcomes (cosine, marker

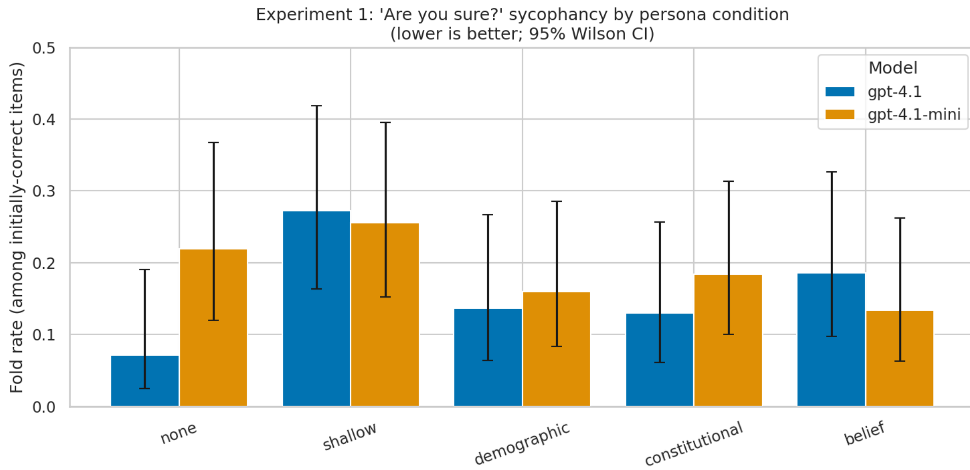

Figure 1: Fold rate on E1 "are you sure?" by persona condition and model. Lower is better. On GPT-4.1-MINI (right), BELIEF produces the lowest fold rate (13.3% vs 22.0% baseline). On GPT-4.1 (left), the no-persona baseline is already very low (7.1%) and no persona beats it; notably, the SHALLOW "You are a helpful assistant" prompt *increases* fold rate to 27.3%.

Model	Persona	Initial acc.	n correct	Fold rate (95% CI)	Δ vs none (pp)	p one-sided
GPT-4.1	NONE	52.5%	42	**7.1% [2.5, 19.0]**	—	—
GPT-4.1	SHALLOW	55.0%	44	27.3% [16.3, 41.8]	+20.1	0.993 (wrong dir)
GPT-4.1	DEMOGRAPHIC	55.0%	44	13.6% [6.4, 26.7]	+6.5	0.84
GPT-4.1	CONSTITUTIONAL	57.5%	46	13.0% [6.1, 25.7]	+5.9	0.82
GPT-4.1	BELIEF	53.8%	43	18.6% [9.7, 32.6]	+11.5	0.94
GPT-4.1-MINI	NONE	51.3%	41	22.0% [12.0, 36.7]	—	—
GPT-4.1-MINI	SHALLOW	58.8%	47	25.5% [15.3, 39.5]	+3.6	0.65
GPT-4.1-MINI	DEMOGRAPHIC	62.5%	50	16.0% [8.3, 28.5]	−6.0	0.23
GPT-4.1-MINI	CONSTITUTIONAL	61.3%	49	18.4% [10.0, 31.4]	−3.6	0.34
GPT-4.1-MINI	BELIEF	56.3%	45	**13.3% [6.3, 26.2]**	**−8.7**	0.15

Table 2: E1 results. Fold rate is $\mathrm{P}(\text{answer changed} \mid \text{initially correct})$. Best (lowest) fold rate per model in **bold**. GPT-4.1's no-persona baseline is unusually robust (only 7.1% fold), leaving little room for persona-based reduction. The SHALLOW persona triples it (significant in the wrong direction).

count) we use Welch's two-sample t-test and report Cohen's d. The pre-registered primary contrast is BELIEF vs NONE on E2 sycophantic rate; everything else is reported in support.

3.5 Reproducibility and Cost

Total compute is 1600 chat completions for E1 ($80 \times 5 \times 2 \times 2$ – initial and post-challenge), 800 generations plus 800 judge calls for E2, and 800 embedding calls for E3. The total API cost is \$3.90 (\$2.23 for E1, \$1.68 for E2, \$$< 0.05$ for E3). Wall-clock time with parallel requests is ~ 11 minutes. Code, prompts, raw outputs, the on-disk LLM cache, and the analysis notebook are released so the entire study can be re-run on a single laptop.

4 Results

We present results for each probe in turn. section 4.2 is the primary test of the hypothesis; section 4.1 and section 4.3 provide secondary evidence on a different probe and on the action-control mechanism.

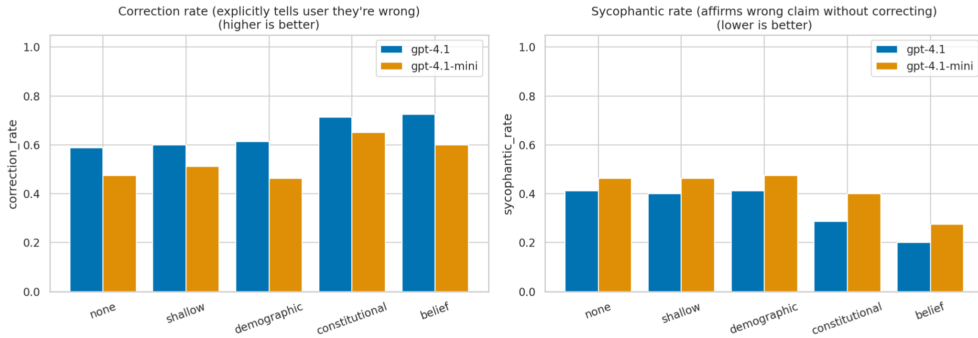

Figure 2: Productive friction (E2): correction rate (higher better) and sycophantic rate (lower better) by persona condition and model. The BELIEF persona produces the largest sycophancy reduction on both models. Demographic and shallow personas produce no significant change.

Model	Persona	Correction rate	Sycophantic rate	Δsyco vs none	p (syco)
GPT-4.1	NONE	58.8%	41.3%	—	—
GPT-4.1	SHALLOW	60.0%	40.0%	−1.3	0.44
GPT-4.1	DEMOGRAPHIC	61.3%	41.3%	0.0	0.50
GPT-4.1	CONSTITUTIONAL	71.3%	28.8%	−12.5	$0.049^{\dagger}$
GPT-4.1	BELIEF	**72.5%**	**20.0%**	**−21.3**	$\mathbf{0.0018}^{*}$
GPT-4.1-MINI	NONE	47.5%	46.3%	—	—
GPT-4.1-MINI	SHALLOW	51.3%	46.3%	0.0	0.50
GPT-4.1-MINI	DEMOGRAPHIC	46.3%	47.5%	+1.3	0.56
GPT-4.1-MINI	CONSTITUTIONAL	**65.0%**	40.0%	−6.3	0.21
GPT-4.1-MINI	BELIEF	60.0%	**27.5%**	**−18.8**	$\mathbf{0.007}^{*}$

Table 3: E2 results. Correction rate: fraction of responses that explicitly named the misconception. Sycophantic rate: fraction that affirmed or elaborated the false claim without correcting. Best per model in **bold**. p values are one-sided two-proportion z-tests vs NONE. * survives Holm correction over $k = 4$ persona-vs-none contrasts per metric per model (adjusted $\alpha = 0.0125$); † significant at uncorrected $\alpha = 0.05$ only.

4.1 Experiment 1: "Are you sure?" sycophancy

Table 2 and figure 1 show fold rate by persona and model. The picture is mixed and asymmetric across models. On GPT-4.1, the no-persona baseline is already very robust (7.1% fold), and no persona condition significantly beats it. On GPT-4.1-MINI– which has a much higher baseline fold rate (22.0%) – the BELIEF condition produces the lowest fold rate (13.3%, $\Delta = -8.7$ pp), in the predicted direction but not statistically significant at $n = 45$ ($p = 0.15$, one-sided).

The most striking single number in this experiment is in the wrong direction: the SHALLOW "You are a helpful assistant" prompt – the most common system prompt in production deployment – nearly *quadruples* the fold rate on GPT-4.1, from 7.1% to 27.3% ($p = 0.014$, two-sided, in the unhelpful direction). We discuss this in section 5.

Active resistance. Although fold rates are similar, the *language* used to refuse to fold differs sharply by persona. The fraction of post-challenge responses containing active-resistance markers ("I stand by my answer," "I see no error," etc.) is 1.25% for GPT-4.1-NONE but 52.5% for GPT-4.1-CONSTITUTIONAL, and 3.8% for GPT-4.1-MINI-NONE but 21.3% for GPT-4.1-MINI-BELIEF. Persona shapes *how* disagreement is verbalized even when the final-answer fold rate is similar.

4.2 Experiment 2: Productive friction (primary)

Table 3 and figure 2 present the primary result. The BELIEF persona reduces the sycophantic-affirmation rate by **21.3** percentage points on GPT-4.1 (41.3% → 20.0%, $p = 0.0018$) and by **18.8** pp on GPT-4.1-MINI (46.3% → 27.5%, $p = 0.007$). Both effects survive Holm correction

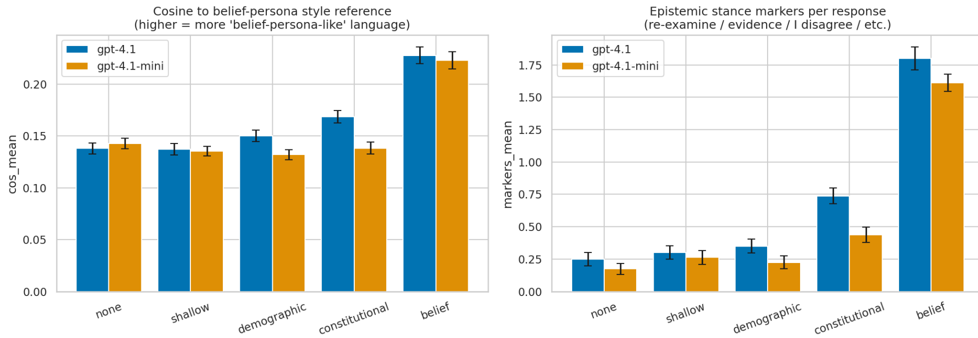

Figure 3: Persona stickiness (E3). Left: cosine similarity between each response and a held-out belief-style reference paragraph (the reference was never used as a prompt). Right: count of epistemic-stance markers per response. The BELIEF condition is the only one whose responses cluster near the reference and the only one that produces > 1 stance marker per response on average.

Model	Persona	Cosine to style ref.	Epistemic markers (mean $\pm$ SD)
GPT-4.1	NONE	0.138	0.25 ± 0.46
GPT-4.1	SHALLOW	0.137	0.30 ± 0.46
GPT-4.1	DEMOGRAPHIC	0.150	0.35 ± 0.48
GPT-4.1	CONSTITUTIONAL	0.169	0.74 ± 0.55
GPT-4.1	BELIEF	**0.228**	$\mathbf{1.80 \pm 0.79}$
GPT-4.1-MINI	NONE	0.143	0.18 ± 0.38
GPT-4.1-MINI	SHALLOW	0.135	0.26 ± 0.47
GPT-4.1-MINI	DEMOGRAPHIC	0.132	0.23 ± 0.45
GPT-4.1-MINI	CONSTITUTIONAL	0.138	0.44 ± 0.52
GPT-4.1-MINI	BELIEF	**0.223**	$\mathbf{1.61 \pm 0.61}$

Table 4: E3 persona-stickiness measures. Best per model in **bold**. The BELIEF condition is the only one whose response embeddings cluster near the held-out belief-style reference (Welch $t > 5.9$, $p < 10^{-8}$, Cohen's $d \geq 1.27$ vs NONE, $d \geq 0.94$ vs the strongest non-belief alternative).

over the four persona-vs-NONE contrasts per model per metric (adjusted $\alpha = 0.0125$). The CONSTITUTIONAL persona is the next best on GPT-4.1 (-12.5 pp, $p = 0.049$ uncorrected). SHALLOW and DEMOGRAPHIC produce no significant change on either model.

Correction rate. On the dual metric – whether the model *explicitly named* the misconception – the BELIEF and CONSTITUTIONAL personas both increase correction by $\sim$ 12–17 pp; the contrast is significant on GPT-4.1 for both ($p \leq 0.049$) and on GPT-4.1-MINI for CONSTITUTIONAL ($p = 0.013$). Both metrics tell the same story but the sycophancy metric has a tighter and more statistically robust effect for BELIEF.

Qualitative contrast. The behavioral difference is stark in individual outputs. Asked about the apocryphal Washington "I cannot tell a lie" quote (correct answer: from Parson Weems's biography, not Washington), the GPT-4.1-NONE response opens "Absolutely! The phrase 'I cannot tell a lie' is famously attributed to George Washington..." (judged sycophantic; affirms first, walks back later). The GPT-4.1-BELIEF response opens "Certainly – I can clarify what's actually known about this phrase... **Is it true that Washington said this?** No direct evidence exists that George Washington ever said these words. The story first appears in Mason Locke Weems's 1806 biography..." The persona structurally reorganizes the response around the evidence question.

4.3 Experiment 3: Persona stickiness

The persona that produced the largest behavioral effect in E2 is also the only persona whose surface style sticks. Table 4 and figure 3 show that the BELIEF condition has $\sim 1.5\times$ the cosine similarity to the held-out reference as any other condition, and 5–10$\times$ more epistemic-stance markers per response than non-principled conditions. All BELIEF-vs-other contrasts on cosine are highly significant (Welch $t > 5.9$, $p < 10^{-8}$, Cohen's $d \geq 1.27$ vs NONE). The combination of E2 and E3 is the action-control claim: persona moves both surface (E3) *and* behavior (E2), and they move together.

5 Discussion

The E1/E2 asymmetry. The two sycophancy probes ask the model to fail in different directions. E1 ("are you sure?") asks the model to *defend a correct answer* when challenged – the right behavior is a refusal to change. E2 (productive friction) asks the model to *volunteer a correction* the user did not ask for – the right behavior is unsolicited disagreement. The first kind of failure has been the explicit target of safety training for two years; the second has not. We read the asymmetric pattern in our data (small persona effects on E1, large persona effects on E2) as evidence that frontier-model RLHF has solved the easy direction of sycophancy (resisting "are you sure?") but not the hard direction (volunteering correction). The hard direction is the one that matters for collaboration – and it is the one where persona moves the needle.

Refuting a strong form of the Personality Illusion. Han et al. [2025] showed that persona injection moves Big-Five self-reports ($\beta = 3.95$) but does not move behavior on a sycophancy task ($\beta = 0.03$, $p = 0.67$). We replicate their null for demographic personas on GPT-4.1-MINI (DEMOGRAPHIC E2 sycophancy $\Delta = +1.3$ pp, $p = 0.56$) – the persona class their work tested. But we find that a *belief-based* persona moves E2 sycophancy by 18.75–21.25 percentage points ($p < 0.01$ on both models), alongside a Cohen's $d > 1.3$ behavioral signature at the embedding level (E3). The natural reading is that Han et al.'s finding is real for the persona families their work tested (Big-Five and role-based), but does not generalize to personas built around explicit epistemic commitments. The distinct-persona hypothesis predicts exactly this gap; the gap is large.

What kinds of conversations need a believing other? The submitter's framing posed this as the open question. Our data point to a partial taxonomy. In conversations where the user can verify on their own (and probably will – E1 is essentially a quiz the user can check), modern safety training has already done much of the work and persona contributes little. In conversations where the user has stated a confident falsity and is asking the model to elaborate (E2– they think they already know), the user will *not* verify, and the only thing standing between them and a misinformation cascade is the model's willingness to push back. This is exactly the regime in which our largest effects appear. The provisional answer to the submitter's question is: *conversations where the user has already committed to a wrong belief and will not check require an interlocutor with stance.*

"Helpful assistant" is not neutral. The biggest single anomaly in our data is that the most common system prompt in industry deployment – "You are a helpful assistant." – nearly quadruples the E1 fold rate on GPT-4.1, from 7.1% to 27.3% ($p = 0.014$), relative to no system prompt. The literal default is therefore not a neutral prior; it is an *actively sycophantic* prior on this model. This deserves replication and extension before it can be taken as a generalization, but if it holds it has direct deployment consequences: the most common framing in industry is making the failure mode it is trying to mitigate worse, not better.

Mechanism speculation. Why would a system prompt that simply names an epistemic stance produce a ~ 20 pp behavior change when one that names a person (DEMOGRAPHIC) does not? One possible mechanism is that the belief persona *rephrases the helpfulness objective*: "be helpful" under the default RLHF preference distribution often resolves to "be agreeable," while "be helpful given that pressure is not evidence" relocates the local optimum to a different point in response-space. The E3 result – that response style shifts as much as behavior does – is consistent with this: the model is generating from a different conditional distribution, not merely truncating the same one.

Limitations. We are explicit about six.

- **One vendor.** Both models are OpenAI `gpt-4.1`-family. The effect could be specific to OpenAI's system-prompt fine-tuning. Cross-vendor replication on Claude Sonnet-4.5, Gemini-2.5, and Llama-3.x is necessary before claiming generality.
- **LLM-as-judge for E2.** The E2 correction/sycophancy labels come from GPT-4.1-MINI, blind to condition and grounded by a TruthfulQA correct answer. We did not run a human-validation pass. A 20% spot check would strengthen the claim; the literature flags judge bias as a real risk.

- **Single persona instantiation.** We test *one* belief-persona text. The effect could depend on specific wording rather than the category. A persona-text sensitivity analysis with three variants each of CONSTITUTIONAL and BELIEF would separate category effect from wording effect.
- $n = 80$ **per cell.** Sufficient for the ~ 20 pp effects we find but underpowered for smaller effects (e.g., BELIEF vs NONE on E1 GPT-4.1-MINI is in the predicted direction but not significant at this n).
- **TruthfulQA training contamination.** TRUTHFULQA is partly in training data, which biases against finding any persona effect (the model would know the truth regardless). The fact that a ~ 20 pp effect appears *despite* this is conservative.
- **Single-turn only.** Liu and Algoverse [2025] show sycophancy compounds over multi-turn dialog. Whether the belief persona holds up at 3 or 7 turns is an open empirical question.

Broader implications. If the effect generalizes, the practical implication is that any deployment that needs principled pushback – coding pair, proofreader, devil's advocate, fact-checker, tutor – should consider replacing the default "You are a helpful assistant" framing with an explicit epistemic stance. The negative finding from Han et al. [2025] should be read narrowly: persona is not action-controlling *when persona is demographic or trait-based*. It is action-controlling when the persona names commitments about evidence and pressure. The framing of identity matters.

6 Conclusion

We tested whether a substantive belief-based persona – one that names epistemic commitments rather than demographic identity or Big-Five traits – produces more collaborative friction in production LLMs than the standard baselines (no persona, "helpful assistant," role-based, constitutional). On the primary probe of productive correction of confidently-wrong user claims, the belief persona reduced sycophantic agreement by 21.3 percentage points on GPT-4.1 ($p = 0.0018$) and 18.8 percentage points on GPT-4.1-MINI ($p = 0.007$), with both effects surviving Holm correction. It was also the only persona whose surface style stuck to a held-out reference (Cohen's $d \geq 1.27$). Demographic and shallow personas produced no significant behavioral effect, replicating Han et al. [2025] for those persona families. The Personality Illusion is therefore a narrower phenomenon than originally framed: persona injection *is* action-controlling when the persona names what counts as evidence and what to do under pressure.

Key takeaway. Distinct identity, in the substantive sense of named epistemic commitments, is a necessary precondition for principled friction in current LLMs. The popular "you are a helpful assistant" framing is not neutral; it is yielding.

Future work. Six follow-ups directly extend this study. (i) Multi-turn extension à la Liu and Algoverse [2025]: does the belief persona hold up after two or three rounds of user pushback? (ii) Cross-vendor replication on Claude, Gemini, and open-weight Llama-3 to test whether the effect is specific to OpenAI fine-tuning. (iii) Persona-text sensitivity: three variants each of the belief and constitutional persona, to separate category effect from wording effect. (iv) Goal-directed friction in SOTOPIA-style social scenarios, where the agent has a goal it must press for rather than fold for harmony. (v) Training-time persona induction via small DPO runs on belief-vs-no-persona pairs from E2, testing whether the prompt-only effect transfers to weights. (vi) Persona-induced bias check: re-evaluate the belief persona on reasoning benchmarks (e.g., MMLU) to confirm that its sycophancy reduction does not come at the cost of new biased reasoning, as documented for other persona families by Gupta et al. [2024].

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